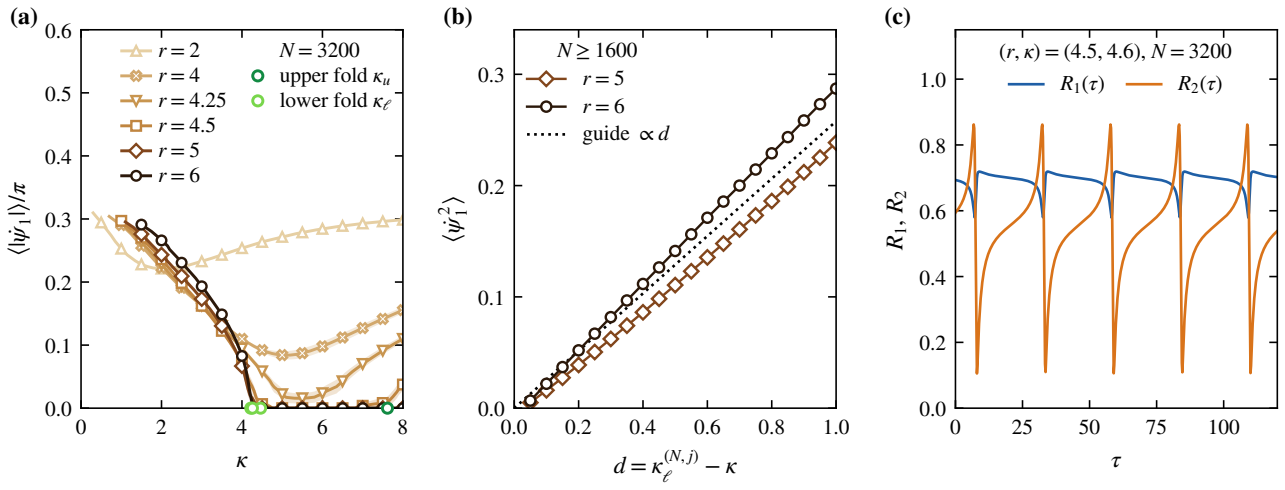
(a)
$N = 3200$
$r = 2$
$r = 4$
$r = 4.25$
$r = 4.5$
$r = 5$
$r = 6$
upper fold κ_u
lower fold κ_ℓ
$\langle|\dot{\psi}_1|\rangle/\pi$
κ
(b)
$N \geq 1600$
$r = 5$
$r = 6$
guide $\propto d$
$\langle\dot{\psi}_1^2\rangle$
$d = \kappa_\ell^{(N,j)} - \kappa$
(c)
$(r, \kappa) = (4.5, 4.6)$, $N = 3200$
$R_1(\tau)$
$R_2(\tau)$
R_1, R_2
τ